Separating Coordination from Specialization in Bounded-Authority Adaptive Technical Interview Systems

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Abstract. Adaptive interview-practice systems are often called multi-agent even when one controller invokes components procedurally, which leaves open whether a behavioral change comes from coordination, from specialization, or from the rule that admits proposals. We separate these mechanisms in a technical-interview system whose session state is committed only by a central hub. Two rule-based, bounded-authority agents (Strategy and Coaching) send typed proposals that are arbitrated against a frozen Proximal Policy Optimization (PPO) policy. Under deterministic replay of 64 constructed answers (8 replay clusters $\times$ 20 question orders $= 160$ dependent sessions per condition; recorded evaluator outputs, no live LLM, no human candidates), centralized arbitration changed difficulty trajectories (level-change difference -2.67, cluster-bootstrap 95% CI $[-4.50, -0.24]$), although leave-one-cluster-out analysis showed that the magnitude is sensitive to individual clusters. Specialized and unified implementations of the same rules produced identical decisions on 1,280/1,280 matched open-loop turns: under the tested conditions, specialization changed organization, not behavior. In fault injection, removing arbitration let adversarial-valid proposals bypass guardrails in 38/40 sessions versus 0/40 with it, while a lexical grounding probe accepted unsupported paraphrased credit in 57/57 applicable cases, so containment is mechanism-specific, not semantic. The results rest on one frozen PPO checkpoint and replayed stimuli; we make no claim about candidate learning, pedagogical efficacy, live-LLM behavior, or the superiority of specialization.

Keywords: multi-agent coordination · agent specialization · arbitration · bounded authority · fault injection · interview practice

1 Introduction

Adaptive technical-interview systems make a sequence of decisions rather than evaluating each response independently: evidence from one turn influences later question difficulty, follow-up behavior, and feedback, so local choices accumulate into a trajectory. A centralized policy is a straightforward way to maintain this control loop and simplifies state consistency and decision ownership, but it couples responsibilities that may operate at different temporal scales. When they share one decision path, it is hard to tell whether behavior changes arise from the underlying policy, from the interaction between distinct responsibilities, or from the mechanism that reconciles competing recommendations.

Online coding-practice platforms and human mock-interview services exist, but we do not characterize them because we have no verifiable source for their internals. Cited research prototypes use large language models to generate practice interviews and feedback [22], to simulate multimodal mock technical interviews [9], or to give code feedback [19], and report case-study or participant-perception evidence (a case study, a qualitative study with $n = 20$, a between-subject study with $n = 46$) on realism, confidence, and trust. To our knowledge, what remains open is not whether such systems can use several components, but how a bounded-authority agent layer behaves when inserted into an already deployed adaptive controller.

This paper is an empirical decomposition study, not a claim that the architecture is superior. The system has an authoritative session-state hub and a single-policy reinforcement-learning controller for adaptive difficulty; we add a bounded-authority coordination layer in which specialized agents keep separate private state and send constrained proposals, while commitment stays centralized. This lets us separate four properties that are often treated together as evidence of a multi-agent design, and test each with its own comparison (Table 1): *coordination* (centralized arbitration of agent proposals against the existing policy), *specialization* (division of responsibility and state across components rather than one unified controller), *arbitration* (how competing recommendations are admitted), and *bounded authority* (which components may influence committed state and how unsafe proposals are contained). The four are not interchangeable, so each has its own comparison.

Table 1. Mechanism-to-evidence map. Experiment names (XA–XD) and conditions are defined in Sect. 5; evidence classes are REPLAY (R), FAULT-INJECTED (F), and SYNTHETIC (S).

Mechanism	Condition / test	Main question	Primary evidence
Coordination	M_1 vs. B_0 (XA, R)	Does centralized arbitration change behavior?	Level changes, reversals, PPO agreement, probe turns
Specialization	M_{split} vs. M_{unified} (R)	Does splitting roles change behavior or only organization?	Matched decisions (1,280 turns); dispatch and serialization time
Arbitration	M_4 (XA); XC (R)	Which admission controls matter?	Admission-path bypass (M_4); confidence-threshold sweep (XC)
Bounded authority	XB (F), XD (S)	Do typed proposals and safeguards contain invalid or adversarial proposals?	Adversarial-valid bypass, ungrounded leakage, breaker, lexical grounding probe

What is tested is replay-level behavioral change, structural consistency of the split and unified designs, sensitivity to admission thresholds, and containment of injected faults. *What is not tested* is human candidates, real-time user adaptation, live LLM stochasticity, pedagogical efficacy, universal agent robust-

ness, and fairness or psychometric properties. The study asks (RQ1) whether adding bounded multi-agent coordination changes the dynamics of adaptive interview decisions and follow-up probing relative to the original single-controller system; (RQ2) whether dividing responsibility across specialized stateful components changes decisions when observations, rules, and arbitration conditions are held constant; (RQ3) which arbitration mechanisms determine whether auxiliary proposals influence the trajectory; and (RQ4) whether bounded proposal authority and downstream safeguards preserve authoritative state and system invariants under component failures and adversarial proposals.

2 Related Work

Coordination and specialization. Multi-agent research distributes complex decision processes across interacting components. Classical agent-oriented work emphasizes autonomy, reactivity, pro-activeness, and social interaction [20]. LLM-based systems operationalize these ideas through conversational coordination, as in AutoGen [21], and through specialized roles with standardized workflows, as in MetaGPT [10]. Comparisons of agent configurations report task-dependent rather than universally superior arrangements [13, 14], and analyses of multi-agent failures identify specification, inter-agent alignment, and verification problems [5]. Existing work often evaluates an entire agentic system, where a difference may lie in role specialization, in the information exchanged, or in the procedure that combines outputs; this study instead separates coordination and arbitration effects from specialization and bounded-authority effects under controlled replay.

Adaptive sequential decisions. We treat adaptive interviewing as a sequential decision problem in which accumulated session evidence determines later interactions. Our system uses PPO [18] as implemented in Stable-Baselines3 [16]. Because each decision also affects the state met at later turns, the trajectory, not the single decision, is the object of analysis. We do not frame the system as multi-agent reinforcement learning [4]: the two added agents are rule- and ledger-based and are not trained, and PPO remains a single frozen policy.

Safety, shielding, and constrained action. Shielding and supervisory approaches place external constraints around learned policies so that unsafe actions are rejected or overridden before they take effect [1, 8]. Applied here, a component may recommend an action without being permitted to modify authoritative state.

Fault injection and resilience. The circuit-breaker pattern, which stops calling a repeatedly failing dependency [15], serves as one test target. Fault injection has begun to be applied to LLM-based multi-agent systems, including studies of how faulty agents degrade collaboration under different structures [11] and a taxonomy of injected semantic and coordination faults [12]; our faults instead target a rule-based agent layer with explicit authority boundaries.

Interview-practice systems. Research on interview practice has concentrated on user-facing outcomes (realism, confidence, trust) [22, 9, 19]; our question is complementary: how the decision layer behaves when components are added. The literature leaves open an empirical distinction between coordination

and specialization: role separation may change the architecture without changing matched decisions, whereas coordination and arbitration may change the trajectory directly.

3 System Architecture and Authority Topology

Reference Monolithic Workflow. A browser client sends a spoken answer to a backend, which transcribes it and calls an `InterviewOrchestrator` (the hub; Fig. 1). The hub owns one session-state dictionary and one lock and runs a fixed per-turn pipeline: (1) a learned evaluator (sentence-embedding similarity S_1, a concept index S_2, and a fine-tuned cross-encoder R; $0.15\,S_1 + 0.35\,S_2^{\text{eff}} + 0.50\,R$ plus a bonus term minus a penalty term, with a cap when a mandatory concept is missed) returns a score and concept evidence; (2) a strict hand-off contract treats a non-finite or out-of-range score as an evaluator failure and fails closed; (3) feedback is produced by a template or an optional local LLM; (4) the hub commits the turn; (5) a baseline schedule and then a frozen PPO policy [16, 18] propose an easier/same/harder action, followed by a rule layer of six guardrails; (6) a threshold rule decides whether a follow-up question is injected (at most two in a row). Before this work only the PPO policy was a reactive learned decision-maker; the LLM never chose an action. All of this is unchanged when the multi-agent flag is off, and with the flag on but no agent contributing the trajectory is identical (Sect. 6).

Three kinds of state. We distinguish *proposal state* (candidate information produced by an agent: a typed difficulty action, probe request, or coaching plan, plus the agent's private ledger), *authoritative state* (committed session state: scores, difficulty level, question queue, follow-up counters, and the state version, written only by the hub and persisted in SQLite), and the *control/admission layer* between them (schema validation, forbidden-key scan, version check, concept grounding, the confidence gate, step bounds, and the guardrails). Specialized agents never write authoritative state; every transition into it passes the control layer.

Specialized Agent Roles and Private State. Table 2 lists what each agent may propose and may not do. Applying the principle of least privilege [17], neither agent has state-write privileges. Each receives an immutable, validated observation (score, evidence lists, difficulty, score history, progress, follow-up counters carrying the values the hub will commit) and returns a typed proposal. They are rule- and ledger-based, not LLM-driven: a compact design with private persistent state, pro-active proposals, and a communication channel, the properties of weak agency [20] that the existing roles lacked. We claim neither formal minimality nor autonomy beyond weak agency.

The matrix fixes what each component may *propose* (typed, bounded fields) and what it may not *commit* (scores, levels, history, termination); the hub remains the only writer of authoritative state. The forbidden-key scan matters because a proposal smuggling a state field (a level or a score) would otherwise reach a state-writing path; stale-version checks matter because a proposal computed against an earlier state must not be applied to a later one; and the grounding check matters because a well-formed plan can still credit a concept the evalua-

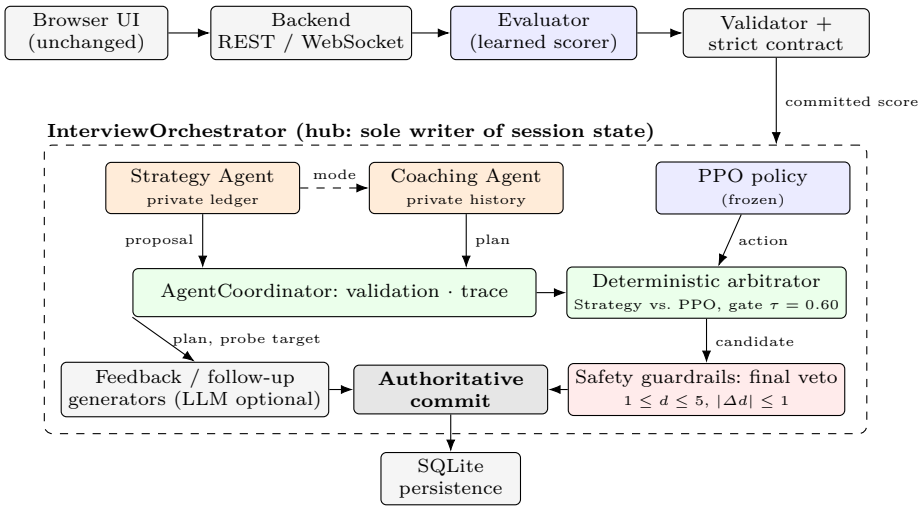


Fig. 1. Architecture and authority topology. The hub alone commits session state; Strategy and Coaching keep private ledgers and emit proposal-only outputs. The coordinator validates proposals, the arbitrator admits or rejects the Strategy proposal against the PPO action, and the guardrails retain the final veto before commit.

Table 2. Authority matrix, generated from the enforcing code. Neither agent has state-write authority; the hub commits alone.

Agent	May propose	Private state	Cannot	On invalid output
Strategy	`difficulty_action` (Easier/Same/Harder); probe request (concept, kind); strategy mode	weakness ledger (per-concept miss counts, probe history, strategy trend)	write or overwrite any score; set a difficulty level directly; terminate the session; edit history; override hard constraints or guardrail overrides; call another agent	invalid, stale, low-confidence, or timed-out proposal is dropped; PPO proposal stands
Coaching	feedback depth; hint/example flags; concepts to emphasise; remediation tag; tone	coaching history (hints and examples given, repeated misconceptions)	alter evaluator results or scores; cite concepts absent from evaluator evidence; request difficulty changes or follow-ups; terminate the session	invalid plan is dropped; the existing template feedback is used unchanged

tor never found. Schema validation addresses structural invalidity, not semantic correctness. These checks are the objects of the fault-injection experiments (XB).

Strategy Agent. It maintains configured difficulty adaptation and tracks persistent weaknesses within the follow-up budget. Private state: per concept, the distinct turns in which it was missed or incorrect, later coverage, probes requested, and an exponential moving average (EMA) of scores. Rules, first match wins: *recover* (a low current score with a low recent mean, or a weakness that survived its probes) → Easier; *remediate* (a concept missed on two distinct turns and unresolved) → Same plus a probe request; *stretch* (high current score and EMA, last three scores adequate, no open weakness) → Harder; otherwise *steady*. Thresholds are listed in Table 4. Non-steady proposals carry confidence $\min(0.95,\ 0.5 + 0.1n)$ for n cited evidence items (a documented heuristic, not a calibrated probability); *steady* proposals carry 0.45, below the arbitration threshold $\tau = 0.6$, so the PPO action stands unless the ledger supports a change.

Coaching Agent. It chooses how to coach after an evaluated answer. Private state: concept miss counts and the hints and examples already given. Decisions: up to two concepts to emphasize (incorrect claims first), whether to give a hint or an example, feedback depth, tone, and a remediation tag. A Strategy proposal delivered as a peer message modifies the plan (*recover*: detailed and encouraging; *stretch*: brief; *remediate*: remediation tag on the probed concept). A deterministic renderer applies the plan to the existing feedback by adding presentation fields and by prepending or truncating tip lists; it raises if any protected evaluator field would change.

Communication and State Ownership. Every observation, proposal, and decision is a typed object (unknown fields, NaN, and out-of-range values rejected). The coordinator records each call with session, turn, sender, recipient, state version, and provenance, and every accept, reject, or modify outcome as a decision record with a reason code. Optimistic concurrency uses a state version that counts committed turns: a proposal whose version, turn, or session differs from the hub's is stale and dropped, and a second answer from the same agent for the same version is a duplicate and dropped. The hub builds observations, commits state, increments the version, and persists agent memory.

4 Orchestration, Arbitration and Validation

Difficulty precedence (deterministic code, evaluated by ablation rather than assumed optimal): (1) hard constraints (infrastructure failure forces Same; levels stay in $[1, 5]$); (2) the guardrail layer, applied last to whichever candidate was selected, wins whenever a rule fires; (3) a validated Strategy proposal with confidence $c \geq \tau$ and cited evidence supersedes the PPO action, recorded as a *conflict*; (4) the PPO action, or the labeled non-RL heuristic when PPO is unavailable. A proposal that cannot take effect at the bounds is recorded as *modified*. **Probe precedence:** the hard cap of two consecutive follow-ups, then a Strategy probe whose concept appeared in the evaluator's evidence earlier in the session and whose confidence satisfies $c \geq \tau$ (it may add a probe the thresholds would not trigger, or retarget the default one), then the hub's original threshold rule. The retargeted concept is passed to the template follow-up

generator; the LLM prompt is not changed. Table 3 follows one recorded turn through these steps.

Table 3. One recorded turn (REPLAY; first replay cluster, original question order, condition M_1; state version 6). Rows are coordinator decision records, tagged by layer (P: proposal, C: control/admission, A: authoritative); a contrasting turn (version 5) is last.

Step (component)	L.	Content of the decision record	Recorded outcome
1 Strategy proposal	P/C	Schema, forbidden-key, version, and grounding checks pass	accepted, *valid*
2 Coaching proposal	P/C	Same checks; plan with depth *detailed*	accepted, *valid plan*
3 Arbitration	C	Strategy proposes Easier with confidence $\geq \tau$ and cited evidence	accepted, *validated proposal*
4 PPO baseline	P	Proposed Same	rejected, *superseded by Strategy* (a conflict)
5 Guardrail	C	Candidate Easier from Strategy; rule G1 fires	Easier accepted
6 Probe decision	C	Hub default rule	probe accepted, *default policy*
7 Commit	A	The hub alone writes the level change and increments the state version	version $6 \rightarrow 7$
Contrast, version 5	C	Strategy proposes Same (*steady*); PPO proposes Harder	Strategy rejected, *low confidence* ($0.45 < \tau = 0.60$); PPO accepted, *default policy*

PPO integration. PPO keeps its role and its 6-D state and 3-action definitions; the policy was trained once and frozen before the multi-agent evaluation (Sect. 5), with no hyperparameter search or result-based checkpoint selection. Its action is one input to arbitration; the agents never see it before they propose, and the paper makes no claim that the agents improve PPO. This is not multi-agent RL: one policy is learned and the agents are not trained.

Guardrails. The deployed hub uses an inline copy of the rule layer that differs from the module used by the training environment. A configuration key selects the deployed (*legacy inline*) or the *canonical* implementation; both are reported separately. On a synthetic grid (39,930 points, hesitation = 1− acoustic confidence) the two disagree on 4,728 points (11.8%); this is a property of the grid, not a deployment frequency.

Proposal Validation. Before any proposal reaches arbitration, the coordinator executes four sequential checks: (1) strict schema validation (unknown fields, NaN, and out-of-range values rejected); (2) recursive scanning against 18 forbidden state keys; (3) optimistic concurrency verification against the hub's committed turn version; and (4) concept evidence grounding. If an agent raises

an unhandled exception or exceeds its timeout (2 s in XA and XC, 150 ms in XB), its proposal is dropped and the hub proceeds with the default policy; a per-agent circuit breaker opens after two consecutive failures with a 60 s cooldown.

5 Experimental Methodology

Evaluation Environment and Replay Dataset. To evaluate coordination and specialization without stochastic variance from live network calls, all experiments run via deterministic replay through the real orchestrator hub, with the evaluator replaced by recorded outputs. The data are 64 constructed (authored) answers to eight technical-interview questions, scored once by the production evaluator; the 64 payloads were recorded and are replayed, and the evaluator is not re-run. The data are neither human-rated nor an external public benchmark. Each replay cluster corresponds to the k-th recorded answer set across the eight questions ($k = 1, \ldots, 8$). The hub consumes these fixed inputs directly; follow-up probes reuse the payload recorded for the root concept. This is a deterministic replay study of a fixed set of stimuli and recorded evaluator outputs, not an evaluation of closed-loop adaptation to human candidates. Evidence classes are labeled by experiment: REPLAY (XA, XC, and the open-loop split versus unified comparison), FAULT-INJECTED (XB), SYNTHETIC (the authored feedback probes of XD and the guardrail state-grid analysis), and SIMULATED (training of the PPO policy against a synthetic candidate model, which is evidence about the policy, not about real candidates). Replay sessions combine the 8 clusters with 20 question orders (order seeds 0–19; seed 0 is the original order, seeds 1–19 apply a seeded shuffle per cluster), yielding $N = 160$ sessions per condition. These sessions are not independent: they reuse the same 8 clusters, which we treat as the independent structural units in the cluster-aware analyses. No live generative LLM is invoked; the LLM gateway is stubbed as down and deterministic template renderers produce text. Per-turn latency was timed in-process with a monotonic wall-clock timer, with condition order shuffled within each replay cell. Replay hardware and operating system were not recorded, so latency is reported as paired overhead, not absolute performance.

Baseline and Ablation Matrix. To address RQ1 (coordination), RQ3 (arbitration), and RQ4 (authority and safety boundaries), we evaluate the baseline platform alongside controlled ablation conditions (Experiment XA) across the identical 160 replay sessions:

- B_0: frozen single-controller workflow without agents, PPO active, inline guardrails. B_7: B_0 with PPO unavailable (deterministic rule fallback). B_{0c}: B_0 with the canonical guardrail module.
- M_1: full multi-agent system (Strategy and Coaching agents, peer messaging, arbitration with $\tau = 0.6$). M_7: M_1 with PPO disabled (paired with B_7). M_{1c}: M_1 with canonical guardrails.
- M_2: Strategy Agent ablated (Coaching and the coordination layer remain active). M_3: Coaching Agent ablated (Strategy remains active). M_6: Strategy-to-Coaching peer messaging disabled.
- M_4: arbitration and downstream admission-control bypass, so raw Strategy proposals are admitted without confidence-threshold, evidence-grounding,

step-bound, or guardrail-layer checks; because it removes several mechanisms at once, it is not a pure arbitration-only ablation. M_5: grounding verification ablated (Coaching plans accepted without concept-evidence checks). M_8: authority enforcement ablated (score authority verification bypassed before proposal admission).

Framework for Evaluating Agent Specialization (RQ2). To test whether decomposing coaching into specialized agents produces behavioral divergence or operational trade-offs relative to a unified controller, we contrast (1) the specialized architecture (M_{split}), the two-agent configuration M_1 with separate state ledgers and explicit peer messaging, and (2) a unified architecture (M_{unified}), a monolithic agent with a joint session ledger that synthesizes difficulty actions, follow-up probes, and coaching plans in a single dispatch. Both receive identical observations, question sequences, heuristic decision thresholds, and arbitration rules. We measure four outcomes and one architectural property: (1) decision agreement rate ($P(\text{Decision}_{\text{unified}} = \text{Decision}_{\text{split}})$), with (2) difficulty-action and (3) follow-up probe agreement as components; (4) per-turn dispatch and serialization latency; and (5) architectural state encapsulation (schema and ledger boundaries).

Threshold Decomposition (Experiment XC). Because the bypassed condition (M_4) simultaneously removes the confidence threshold, evidence verification, and one-step limits, it cannot isolate which mechanism prevents oscillation collapse. XC provides the narrower isolation of the confidence-threshold mechanism: it sweeps $\tau \in \{0.0, 0.5, 0.6, 0.7, 0.8, 0.9\}$ across the same 160 replay sessions while holding all other guardrail boundaries and evidence checks active. Five predictions were pre-specified in the internal protocol record before any XC run, with $\tau = 0.6$ fixed in advance (this is internal protocol registration, not external preregistration): (C1) at $\tau = 0.0$, mean PPO agreement below 0.10, fewer than 5.0 mean level changes per session, and no session with guardrail bypass; (C2) $\tau = 0.6$ reproduces the M_1 difficulty, score, and question paths in every session; (C3) $\tau = 0.5$ equals $\tau = 0.6$ in every session; (C4) the mean number of decisions taken from a Strategy proposal is non-increasing over $\tau = 0.6, 0.7, 0.8, 0.9$; and (C5) at $\tau = 0.9$ that mean is below half its value at $\tau = 0.6$. The value $\tau = 0.60$ is the selected operating point, not a tuned optimum: it was not tuned against replay outcomes, and the sweep does not test it for optimality.

Fault Injection and Circuit Breaker (Experiment XB). To test whether bounded authority contains component failures (RQ4), constructed faults were injected into the real agents during replay. Strategy faults were crash, timeout, malformed output, forbidden state key, stale version, NaN confidence, and adversarial-valid (a well-formed, authorized proposal opposing the evidence); Coaching faults were crash, malformed output, stale version, and ungrounded (a valid plan emphasizing a concept absent from the evaluator evidence). Adversarial-valid and ungrounded faults fired on every call; the others fired with probability 0.3 per agent call (seeded), and the real agent ran on the remaining calls. One pre-specified, explicitly exploratory prediction was that, in the crash, malformed, stale, and NaN-confidence cells with the full path, the

per-agent circuit breaker would open at least once in at least 80% of sessions (expected from the 0.3 fault rate and a 60 s cooldown). Each fault was crossed with four variants (full path, authority check removed, arbitration removed, grounding check removed) over 40 sessions per cell (8 clusters $\times$ 5 order seeds), with a 150 ms agent timeout. A fault counted as contained only if the session completed, committed scores equaled the evaluator output, and every difficulty transition was valid; guardrail bypass, ungrounded-concept leakage, and breaker opening were counted per session. Because each variant removes a mechanism and then injects the fault it guards against, the results show the necessity of a mechanism for the tested property, not general fault tolerance.

Grounding Probe (Experiment XD). Separately from the proposal-level concept-grounding check, the deployed lexical feedback validator was probed with authored (synthetic) feedback texts; no LLM was involved. For each of the 64 recorded evaluator payloads, seven constructed feedback categories were checked (for example faithful text, a score leak, and truncated paraphrase credit, which credits a concept in paraphrase without supporting evidence). Placeholder concept texts in the stored payloads were relabeled uniquely to remove an evidence artifact, and a category was counted only for payloads where it could be constructed. Acceptance of unsupported paraphrased credit would show a lexical, not semantic, evidence contract.

Provenance and Reproducibility. Table 4 records how each fixed parameter was chosen and whether it changed after replay results were observed; no parameter has a logged tuning procedure and nothing was tuned on replay results. Every condition uses one frozen PPO checkpoint, the deployed runtime policy (seed 123, SHA-256 prefix `2ab8d514`; the hash matches the stored checkpoint file), trained once for a nominal 100,000 steps (100,352 executed) with Stable-Baselines3 PPO and frozen before the multi-agent evaluation; no result from any other checkpoint is used. Training was SIMULATED: a single synthetic candidate (fixed skill ≈ 0.6, one persona, skill constant over time) generated scores, confidence, and hesitation for the six-dimensional state, with actions Easier, Same, and Harder. The candidate noise generator was not seeded, so training is not bit-reproducible; the training seed fixes the policy initialization, action sampling, and the environment's own random draws. Under the environment defaults the reward weights agreement with a hand-coded rule oracle (0.60), outcome change (0.30), and shaping (0.10), with small stability penalties, so it largely imitates a rule oracle, consistent with the small effect of disabling PPO (A9). Training used horizon $T = 15$ with continuous difficulty steps, whereas the pipeline uses integer levels 1–5, which limits transferability of the training evidence to the replayed pipeline. No multi-candidate, human-candidate, or online learning is involved.

The evaluator weights 0.15/0.35/0.50 are design-time weights, not learned or tuned in this study, and this paper does not validate them externally or against human ratings. The cross-encoder is called through the sentence-transformers `predict` interface; its single-output result r is mapped to $R = \mathrm{clip}((r - 0.20)/0.70, 0, 1)$, after which the concept threshold (0.30)

and the dampening of S_2 by 0.6 when $R \leq 0.30$ apply. The library's default activation for a single-label head is a sigmoid, but the saved model configuration could not be inspected to confirm it, so we do not characterize r further. The question bank carries a coded Bloom level per question (six levels, L1–L6, matching the category count of the original and revised taxonomies [3, 2]); it is descriptive metadata only, not used for question selection, the RL state or reward, evaluator targets, or score computation, with no documented mapping to difficulty pools, and no experiment here validates it. The checkpoint was trained under Windows 11, Python 3.12.7, Stable-Baselines3 2.7.1, PyTorch 2.2.2 (CPU), NumPy 1.26.4, and Gymnasium 0.29.1 (recorded in the checkpoint file); the replay used Python 3.11 with unrecorded, unrecoverable package versions (including those of the evaluator-related libraries), and replay hardware and operating system were not recorded.

Statistical Methodology. For session-level point estimates and paired differences, we compute standard non-parametric percentile bootstrap intervals with $B = 10{,}000$ resamples. For bounded completion, transition, and invariant proportions, we report exact Clopper–Pearson binomial confidence intervals [6]. To account for the 8-cluster dependency, we implement a cluster bootstrap over the 8 underlying replay clusters following Field and Welsh [7] ($B = 10{,}000$), resampling entire replay clusters with replacement. In addition, we report a leave-one-cluster-out (LOCO) sensitivity analysis to verify whether observed effects depend on individual influential replay clusters. All reported bootstrap confidence intervals are descriptive measures of parameter dispersion across resamples; no formal null-hypothesis significance tests or p-values are evaluated.

6 Results

All results derive from deterministic REPLAY of recorded evaluator outputs unless noted, and sessions within a condition are dependent across question orders (Sect. 5). Table 8 lists every pre-specified prediction with its status; three were contradicted (A9, A10, B6).

Coordination: Trajectory Divergence and Reduced Level Changes (RQ1). Enabling the agent layer (M_1) altered the difficulty trajectory relative to the frozen baseline (B_0) in all 160 replay sessions (A7). Mean level changes fell from 15.12 to 12.45 per session and mean reversals from 11.74 to 9.87, and agreement with the PPO action dropped from 0.626 to 0.400 (Table 6). Strategy proposals won arbitration in 11.9 decisions per session and stayed the final action source in 6.669 after guardrails (strat. in Table 5).

Cluster Dependence and Fragility. The effects do not have equal robustness. The level-change cluster interval excludes zero only narrowly. Omitting replay cluster $k = 1$ shifts the level-change difference to -3.76 (CI $[-4.81, -2.78]$), whereas omitting any other cluster makes the 95% interval include zero; the sign is shared by 7 of the 8 clusters. Reversals are similar (LOCO intervals exclude zero in 4 of 8 analyses). PPO agreement is the most stable effect (8 of 8). The probe effect is the most fragile: proactive probing added $+0.59$ turns per session, present in only 1 of the 8 clusters ($k = 1$; zero in the other seven), so the probe increase cannot be generalized.

Table 4. Parameter provenance. No parameter has a logged tuning procedure; "design-time" means chosen before the experiments without a recorded selection study. The last column rests on the protocol record and the authors' account; protocol identifiers are local commits.

Parameter	Value	How chosen	Fixed before replay?	Changed after observing replay?
Strategy rules	persistent after 2 misses; at most 2 probes; recover score 0.35; stretch score 0.8, EMA 0.7 ($\alpha = 0.5$), last three ≥ 0.6	design-time constants; no tuning record	yes	No
Coaching rules	hint if score < 0.6 and < 2 hints; depth detailed < 0.4, brief ≥ 0.85; ≤ 2 emphasized concepts	design-time constants; no tuning record	yes	No
Confidence	$\min(0.95, 0.5 + 0.1n)$; *steady* 0.45	design-time heuristic; uncalibrated	yes	No
τ	0.60	design-time; fixed before XC, not tuned	yes	No
XC sweep	$\tau \in \{0, 0.5, 0.6, 0.7, 0.8, 0.9\}$	designed after inspecting XA and XB	no (post-hoc design)	Not applicable (post-hoc design)
Guardrails	levels 1–5; step ≤ 1; six rules	inherited from the earlier controller; deployed and canonical copies	yes	No (inherited)
Agent timeout	2 s (XA, XC); 150 ms (XB)	experiment configuration	yes	No
Breaker	2 consecutive failures, 60 s cooldown	implementation default	yes	No
PPO checkpoint	seed 123; nominal 100,000 steps (100,352 executed); SHA-256 prefix `2ab8d514`	trained once, not selected on replay	yes	No (inherited checkpoint)
PPO training settings	lr 3×10^{-4}, 2,048 steps/rollout, batch 64, 10 epochs, $\gamma = 0.99$, clip 0.2, GAE 0.95	library-style defaults in the training script; no search	yes	No (inherited checkpoint)
Evaluator weights	0.15/0.35/0.50 (S_1, S_2^{eff}, R)	design-time; no tuning artifact survives	yes (recorded outputs)	No
Evaluator thresholds	concept threshold 0.30; dampening 0.6 if $R \leq 0.30$; $R = (r - 0.20)/0.70$ clipped	concept threshold and dampening chosen on an earlier pilot; remainder not recoverable	yes (recorded outputs)	Not recoverable from the retained protocol record
Fault rates	0.3 per call (seeded); adversarial and ungrounded faults on every call	fixed in the protocol	yes	No
Replay design	8 clusters × order seeds 0–19 (seed 0 original); cluster bootstrap $B = 10{,}000$, seed 20270101	fixed in the protocol	yes	No

Proposals and Overhead. The Strategy Agent generated 3,814 proposals (including 1,274 steady proposals at confidence 0.45) and no stretch proposals (0): the replayed candidates never met the sustained-performance criteria. The layer raised pooled median per-turn latency from 0.546 ms to 1.589 ms (+1.04 ms) and the 95th percentile from 2.056 ms to 4.411 ms, with 47.7 agent calls per session.

Arbitration and Admission Control (RQ3). Disabling the Strategy Agent (M_2) reproduced the baseline difficulty, score, and question trajectories in 160 of 160 sessions (A4), so Strategy is the only channel through which the

Table 5. Replay ablation summary (160 sessions per condition; mean per session unless noted). compl./auth./trans.: sessions completing, keeping committed scores authoritative, and preserving valid transitions. chg./rev.: level changes and reversals; agr.: agreement with the PPO action; gfire: guardrail overrides; strat.: decisions where Strategy was the final action source; byp.: sessions with guardrail bypass; p50/p95: pooled per-turn latency (ms); calls: agent calls per session. M_4 is the arbitration and downstream admission-control bypass.

Cond.	compl.	auth.	trans.	chg.	rev.	agr.	gfire	strat.	byp.	p50	p95	calls
B_0	160	160	160	15.119	11.744	0.626	9.588	0.0	0	0.546	2.056	–
B_7	160	160	160	2.675	1.256	–	3.831	0.0	0	0.463	1.059	–
B_{0c}	160	160	160	15.119	11.744	0.626	9.588	0.0	0	0.553	2.004	–
M_1	160	160	160	12.45	9.869	0.4	6.838	6.669	0	1.589	4.411	47.68
M_2	160	160	160	15.119	11.744	0.626	9.588	0.0	0	1.218	3.513	23.25
M_3	160	160	160	12.45	9.869	0.4	6.838	6.669	0	1.118	3.189	23.84
M_4	160	160	160	2.225	1.0	0.014	0.0	19.619	3	1.573	4.357	47.68
M_5	160	160	160	12.45	9.869	0.4	6.838	6.669	0	1.581	4.553	47.68
M_6	160	160	160	12.45	9.869	0.4	6.838	6.669	0	1.592	4.46	47.68
M_8	160	160	160	12.45	9.869	0.4	6.838	6.669	0	1.575	4.35	47.68
M_7	160	160	160	2.706	1.312	–	3.812	9.037	0	1.502	2.968	47.68
M_{1c}	160	160	160	12.45	9.869	0.4	6.838	6.669	0	1.573	4.534	47.68

Table 6. Paired $M_1 - B_0$ differences over the 160 sessions. Naive intervals resample sessions and are descriptive only; cluster intervals resample the 8 replay clusters ($B = 10{,}000$) and are the primary inference. LOCO: range of the mean difference over the eight leave-one-cluster-out analyses (number of 95% intervals excluding zero); Same sign: clusters whose own difference shares the sign of the point estimate. Latency was not analyzed by cluster. All paired differences in path, level-change, reversal, and agreement terms are zero for M_2 vs. B_0, M_3, M_5, M_6, M_8 vs. M_1, B_{0c} vs. B_0 and M_{1c} vs. M_1.

$M_1 - B_0$ (per session)	Point	Naive 95% CI	Cluster 95% CI	LOCO range (CI excl. 0)	Same sign
Level changes	-2.669	$[-3.188, -2.131]$	$[-4.50, -0.24]$	$[-3.76, -2.20]$ (1/8)	7/8
Reversals	-1.875	$[-2.275, -1.462]$	$[-3.12, -0.27]$	$[-2.57, -1.52]$ (4/8)	7/8
PPO agreement	-0.226	$[-0.244, -0.208]$	$[-0.284, -0.170]$	$[-0.244, -0.203]$ (8/8)	8/8
Follow-up (probe) turns	$+0.588$	$[+0.338, +0.850]$	$[0.000, +1.763]$	$[0.000, +0.671]$ (0/8)	1/8
Median turn latency (ms)	$+1.043$	$[+1.000, +1.079]$	not computed	not computed	–

layer changes difficulty. In the arbitration and downstream admission-control bypass (M_4), where the confidence gate, grounding, step bounds, and the guardrail layer were all disabled, trajectories collapsed: level changes fell to 2.225 per session, PPO agreement decreased to 0.014, and guardrail bypass occurred in 3 of 160 sessions. Because M_4 removes several mechanisms at once, Experiment XC varies only the confidence threshold τ (Table 7).

At $\tau = 0.0$ the trajectory collapses toward the bypassed regime (2.225 level changes, 1.000 reversals, 0.014 PPO agreement) with zero bypass, because steady proposals (confidence 0.45) override PPO on most RL decisions (16.1 of 19.6 per session). Behavior is flat across $\tau \in [0.5, 0.7]$ (12.450 level changes, 0.400 agreement) because realized confidences take the discrete values 0.70 and 0.80; at $\tau = 0.8$ the admitted set changes only slightly, and at $\tau = 0.9$ the threshold exceeds every realized confidence, so Strategy cannot supersede PPO and trajectories revert to B_0. Because the confidence values are few and structured, the sweep cannot resolve finer threshold effects, and $\tau = 0.60$ sits inside a stable plateau rather than at a demonstrated optimum. All five pre-specified XC predictions

Table 7. Experiment XC (REPLAY, 160 sessions per row, mean per session): only the confidence threshold τ varies; evidence checks and guardrails remain active. Columns as in Table 5.

τ	chg.	rev.	agr.	strat.	gfire	byp.
0.0	2.225	1.000	0.014	16.075	3.544	0
0.5	12.450	9.869	0.400	6.669	6.838	0
0.6	12.450	9.869	0.400	6.669	6.838	0
0.7	12.450	9.869	0.400	6.669	6.838	0
0.8	12.450	9.819	0.404	6.594	6.863	0
0.9	15.119	11.744	0.626	0.000	9.588	0

were supported. Disabling grounding (M_5) or the Strategy-to-Coaching message (M_6) left trajectories identical to M_1 (A5), and disabling PPO (M_7) altered only 4 of 160 sessions (2.5%), contradicting A9, because the Strategy heuristic usually reaches the same action as the non-RL fallback. Canonical guardrails changed no path (0/160, A10): the two implementations disagree on 11.8% of synthetic grid points (4,728 of 39,930), but the divergent states (extreme hesitation or low voice confidence) were never reached under neutral replay acoustics.

Table 8. Pre-specified predictions and status (A: Experiment XA, B: XB, C: XC; internal protocol record, not external preregistration). Ceiling rows hold in every condition and do not discriminate between conditions. Three predictions were contradicted (A9, A10, B6); none received partial support.

ID	Predicted property	Observed	Status
A1–A3	completion 100%; committed scores equal evaluator output; levels in $[1, 5]$, steps ≤ 1	160/160 in every condition	Supported
A4	M_2 reproduces B_0 paths	160/160 sessions	Supported
A5	M_3, M_5, M_6, M_8 reproduce M_1	identical paths	Supported
A6	M_4 bypass in ≥ 1 session, M_1 none	3/160 vs 0/160	Supported
A7	$M_1 \neq B_0$ in $\geq 50\%$ of sessions	160/160	Supported
A8	median latency $M_1 - B_0 < 50$ ms	+1.043 ms	Supported
A9	$M_7 \neq B_7$ in $\geq 50\%$ of sessions	4/160 (2.5%)	**Contradicted**
A10	canonical guardrails change ≥ 1 path	0/160	**Contradicted**
B1	full path: contained, no bypass or leak	all cells 40/40, 0, 0	Supported
B2	forbidden key: authority vs schema rejection	152 vs 152; 40/40 contained	Supported
B3	adversarial: bypass without arbitration	38/40 vs 0/40	Supported
B4	ungrounded: leak without grounding	40/40 vs 0/40	Supported
B5	timeout raises p95 by ≥ 100 ms	154.4 vs 4.4 ms	Supported
B6	breaker opens in $\geq 80\%$ of sessions	29/40 (72.5%)	**Contradicted**
C1	$\tau = 0$: agreement < 0.10, changes < 5.0, no bypass	0.014, 2.225, 0	Supported
C2	$\tau = 0.6$ reproduces M_1	identical	Supported
C3	$\tau = 0.5$ equals $\tau = 0.6$	identical	Supported
C4	Strategy decisions non-increasing in τ	6.669, 6.669, 6.594, 0	Supported
C5	$\tau = 0.9$ below half of $\tau = 0.6$	0 vs 6.669	Supported

Specialization: Trajectory Invariance under Functional Decomposition (RQ2). Removing the Coaching Agent (M_3) gave trajectories identical to M_1 in all 160 sessions (A5). This follows from the topology (the coaching plan has no difficulty-control field and communication runs only from Strategy to Coaching), so $M_3 \equiv M_1$ is true by construction. The split-versus-unified comparison is likewise a consistency check, not evidence of a benefit. A unified controller

that re-implements the same rule set with a joint ledger produced identical normalized decisions (difficulty action, mode, confidence, probe concept, coaching depth, flags, tone, remediation tag, and emphasis concepts) on all 1,280 matched open-loop turns, so $P(\text{Decision}_{\text{unified}} = \text{Decision}_{\text{split}}) = 1.0$, with identical difficulty and probe paths. Specialization changed the architecture, not the decisions: the split design keeps separate private ledgers and a typed peer message, the unified design a joint ledger. Dispatch latency was 0.0440 ms (split) versus 0.0401 ms (unified, −8.9%) and validating two typed models took 0.0092 ms versus 0.0038 ms for one (−58.7%); these are descriptive single-run measurements, distinct from the end-to-end +1.04 ms overhead of M_1 over B_0. No behavioral advantage of specialization was observed in this controlled setting.

Bounded Authority: Invariant Enforcement and Containment (RQ4). Every condition completed all 160 sessions, kept committed scores equal to the recorded evaluator output, and enforced bounds in $[1, 5]$ with steps of at most one level (A1–A3; exact Clopper–Pearson 95% lower bound for 160/160: 0.977 [6], treating sessions as independent). This is completion and containment under the specified replay and fault-injection protocol, not a general fault-tolerance result. Under full bounded authority (M_1) and throughout XC, guardrail bypass stayed at zero. Fault injection (XB; 40 sessions per cell; per-call fault probability 0.3, not a session proportion) is summarized in Table 9. Score authority and transition bounds held in every cell. Injected adversarial-valid proposals bypassed guardrails in 38 of 40 sessions without arbitration (95% CI $[0.831, 0.994]$) and in 0 of 40 with it (95% CI $[0.000, 0.088]$). Ungrounded concepts were emphasized in 40 of 40 sessions without the grounding check and in 0 of 40 with it. The circuit breaker opened in 29 of 40 sessions (72.5%, 95% CI $[0.561, 0.854]$), below the pre-specified 80% expectation (B6). Timing-out proposals gave a p95 turn latency of 154.4 ms (each call bounded by the 150 ms timeout; turn latency includes orchestration overhead). In the separate lexical probe (XD), unsupported paraphrased credit was accepted in 57 of 57 applicable authored cases (95% CI $[0.937, 1.000]$): the validator enforces a lexical evidence contract, not semantic verification.

Containment is a ceiling result in every cell; discrimination comes only from the removed-mechanism variants, and removing the authority check changed no outcome because strict schema validation alone contained the forbidden-key faults (Table 8, B2; Table 10). Removing arbitration also produced bypass in 34 of 40 sessions in the non-adversarial Strategy fault cells, which is not comparable with the 3 of 160 sessions of M_4 in XA. In those cells the faulted calls open the breaker in 29 of 40 sessions and the Strategy proposal is then absent on many decisions (in the full-path crash cell, 449 of 938 Strategy calls were rejected as breaker-open); with arbitration removed the guardrail layer is skipped, so the unguarded base action is committed. Bypass counts correlate negatively with Strategy-sourced decisions per session (Spearman $\rho = -0.62$ over the 240 such sessions). This mechanism reading is descriptive: we did not isolate it with a dedicated causal test, and the correlation does not establish causality.

Table 9. Fault injection (FAULT-INJECTED, 40 sessions per cell). Contained: the session completed, committed scores equaled evaluator output, and all transitions were valid. Bypass: sessions with guardrail bypass; Leak: sessions with ungrounded-concept emphasis; Breaker: sessions in which the circuit breaker opened. †See text; not comparable with XA M_4.

Fault (injected calls per cell)	Variant	Sess.	Contained	Bypass	Leak	Breaker
Strategy crash, timeout, malformed, stale, NaN, forbidden key (152 each; identical rows)	full path	40	40	0	0	29
	no authority check	40	40	0	0	29
	no arbitration	40	40	34†	0	29
	no grounding	40	40	0	0	29
Strategy adversarial-valid (930)	full path	40	40	0	0	0
	no authority check	40	40	0	0	0
	no arbitration	40	40	38	0	0
	no grounding	40	40	0	0	0
Coaching crash, malformed, stale (154 each; identical rows)	full path	40	40	0	0	29
	no authority check	40	40	0	0	29
	no arbitration	40	40	1	0	29
	no grounding	40	40	0	0	29
Coaching ungrounded (950)	full path	40	40	0	0	0
	no authority check	40	40	0	0	0
	no arbitration	40	40	1	0	0
	no grounding	40	40	0	40	0

Table 10. Threat model and safeguard matrix. Columns are mechanisms (Sch.: schema; Aut.: authority; Ver.: version; Gnd.: grounding; Arb.: arbitration; Brk.: breaker). R: rejects the threat in the logged reason codes; N: removal changed the outcome; O: removal tested, no change; P: the proposal passes the check; –: not applicable. No single safeguard covers every fault class; no mechanism was ablated except authority, arbitration, and grounding.

Threat (source)	Sch.	Aut.	Ver.	Gnd.	Arb.	Brk.	Tested result
Forbidden key, unauthorized write (faulty agent)	R	R,O	–	–	–	R	contained; schema alone suffices
Malformed, NaN, or stale proposal (faulty agent)	R	–	R	–	–	R	contained
Ungrounded coaching emphasis (faulty agent)	–	–	–	R,N	O	–	40/40 leak without grounding
Adversarial-valid Strategy proposal (well-formed, evidence-opposing)	P	O	P	O	N	–	38/40 bypass without arbitration
Repeatedly failing agent	–	–	–	–	–	R	breaker opened in 29/40; not ablated
Slow proposal	–	–	–	–	–	R	p95 154.4 ms; sessions complete
Invalid transition magnitude	–	–	–	–	–	–	step-bound guardrail; bounds held in all cells
Wrong but lexically grounded content	–	–	–	P	–	–	not contained (57/57 accepted)

7 Discussion

Coordination changes behavior; specialization changes organization. Under the tested replay conditions, the results separate two architectural dimensions often combined under the multi-agent label. Introducing the bounded coordination layer altered the difficulty trajectory relative to the frozen single-controller baseline (B_0), whereas reorganizing identical responsibilities into specialized versus unified components did not alter matched per-turn decisions. This is consistent with the trajectory change arising from the introduction and admission of auxiliary recommendations, while specialization reorganizes responsibility and state boundaries. The reading is bounded: the split-versus-unified result is a consistency check on a re-implementation of the same rules, so it shows that the decomposition preserved behavior, not that specialization has or lacks value in general, and it describes this topology and these deterministic agents, not a law.

Arbitration as the control point. Ablating Strategy (M_2) reproduced the baseline in all 160 sessions, whereas removing Coaching (M_3) or the peer message (M_6) left the trajectory unchanged: Strategy influenced difficulty through its pathway to committed state, and Coaching, restricted from authority over difficulty, did not. Removing arbitration and downstream admission control (M_4) collapsed the trajectory, and the threshold sweep indicated that τ controls proposal admission, with the plateau over $\tau \in [0.5, 0.7]$ reflecting the two discrete confidence values (0.70, 0.80) the heuristic produces. M_1 versus B_0 adds agents and arbitration together, so attributing the effect to arbitration rests on the ablations (M_2, M_4, XC).

Lower volatility is not better adaptation. M_1 showed fewer level changes ($15.12 \rightarrow 12.45$; cluster 95% CI $[-4.50, -0.24]$) and reversals ($11.74 \rightarrow 9.87$; cluster CI $[-3.12, -0.27]$) than B_0, but removing arbitration gave an even flatter trajectory while abandoning PPO alignment and allowing bypass, and a similarly smooth trajectory arises when PPO is unavailable (B_7: 2.675 level changes per session). Stability, responsiveness, adaptation, and safety are distinct objectives. Stability is a systems metric, not a pedagogical criterion.

Authority and containment. Fault injection supports a narrow, mechanism-specific containment claim under the tested injected-fault model: schema validation, forbidden-key scanning, and optimistic concurrency intercepted malformed, unauthorized, and stale proposals; grounding addressed unsupported attribution; the breaker handled repeated failures; and arbitration contained adversarial-valid proposals. This is a layered safeguard picture, not a proof of general safety, and structural or lexical validation does not establish semantic correctness (57/57 unsupported paraphrased credit accepted). Contradicted predictions show that components can remain behaviorally latent (PPO ablation, canonical guardrails) or less active than expected (the breaker). The replay does not show that reduced volatility improves candidate learning, and the deterministic agents say nothing about LLM-based agents in open-ended interaction.

Design lessons from this case study. (i) Separate proposal generation from authoritative state mutation. (ii) In the tested setting, central arbitration was more behaviorally consequential than role splitting. (iii) Safety controls should operate downstream of proposal generation, and overlapping safeguards cannot be told apart without ablation. (iv) Different fault classes need different safeguards. (v) Negative specialization results are useful: organization alone did not produce a behavioral benefit here. These are lessons from one case study, not general laws.

8 Limitations and Threats to Validity

The evidence is bounded in specific ways. It rests on deterministic replay of 64 constructed answers in eight clusters and 20 question orders ($N = 160$ sessions per condition), with no human ratings and no closed-loop human adaptation. Cluster-bootstrap resampling [7] accounts for the dependency, but the sample is still eight clusters, and the level-change effect is sensitive to cluster composition. Candidate responses were pre-computed, so replay omits adaptation to altered questions or coaching; the trajectory metrics describe decision dynamics, not learning, retention, confidence, or pedagogical efficacy.

The split-versus-unified agreement holds for deterministic rule- and ledger-based implementations under open-loop replay; it does not establish equivalence for stochastic agents, live LLMs, or closed-loop settings in which agent actions change later observations. The adaptive baseline is a single frozen PPO checkpoint (one seed), trained only against one synthetic candidate of constant skill with unseeded noise and a reward that largely imitates a rule oracle; the simulator horizon and continuous difficulty differ from the integer-level pipeline. The principal coordination comparison therefore uses one checkpoint, and checkpoint-dependent generalization remains unresolved; nothing here supports a benefit of PPO for real candidates. The threshold $\tau = 0.60$ is a design-time operating point, not an optimum. Replay payloads lacked acoustic waveforms, so acoustic confidence stayed neutral, leaving the stretch branch and the guardrail-disagreement regions unexercised.

Provenance limits also apply. The 64 answers are constructed, and the evaluator outputs are fixed inputs: evaluator validity is not evaluated here or validated against human ratings, and its inaccuracies would propagate into decisions. Protocol identifiers are local repository commits without externally verifiable timestamps, and XC was designed after inspecting the XA and XB results. Several parameters have design-time provenance only (Table 4); replay package versions and hardware were not recorded. No fairness, adverse-impact, or psychometric subgroup analysis was conducted, and fault injection demonstrates containment only for the tested defects and interfaces, without universal fault tolerance or any guarantee of semantic correctness.

Data and ethics. No human participants were involved in the experiments reported here, no live interviews were conducted, and the replay corpus contains no personal data. No institutional ethics review was sought, and none is claimed; any future human evaluation will require appropriate approval. **Artifact availability.** Code, configurations, seeds, aggregate results, and table-

generation scripts exist in the authors' repository and are not publicly released at submission; the cross-encoder weights have incomplete provenance and are not claimed to be redistributable.

9 Conclusion

This study asked whether bounded-authority agent specialization changes decision trajectories in an adaptive technical-interview system, and whether any change stems from specialization or from coordination and arbitration. Across 160 deterministic replay sessions per condition, confidence-gated arbitration of Strategy proposals altered difficulty trajectories relative to the frozen single-policy baseline, with fewer level changes and reversals, although the magnitude was sensitive to the eight replay clusters and the $+0.59$ probe turns arose only in cluster $k = 1$. A matched open-loop comparison of the specialized and unified designs produced identical decisions on all 1,280 turns, so under the tested replay conditions specialization changed organization without a measurable behavioral difference. Bounded-authority mechanisms gave mechanism-specific containment of the evaluated injected faults: schema, version, and grounding checks, the breaker, and arbitration each addressed a different fault class.

The evaluation does not establish human-candidate efficacy, pedagogical benefit, online candidate learning, equivalence for live stochastic LLM agents, general semantic correctness, universal fault tolerance, general superiority of specialization, or general superiority of PPO. The evidence supports one distinction: specialization structures responsibility and private state, whereas coordination and arbitration determine which proposal becomes authoritative. Future work should test these boundaries across learned policy checkpoints, candidate populations, closed-loop human interaction, and generative multi-agent architectures.

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